

Artificial Intelligence and Computer Algebra Systems: What Role in Mathematics Education?

Michel Beaudin¹

¹ École de technologie supérieure, Montréal, Québec, Canada
michel.beaudin@etsmtl.ca

It would be presumptuous, to say the least, to claim that computer algebra systems had already predicted the advent of artificial intelligence. Given the recent advances in AI - particularly in mathematics - and its far broader scope than that of symbolic calculators, the comparison is not entirely appropriate.

Nevertheless, several colleagues are already examining the effects of artificial intelligence on our teaching - is it still relevant to assign homework? - and the upheavals ahead may well prove difficult to manage. Even so, we should not forget the significant impact that the emergence of Maple, Mathematica, Derive, and similar systems had on mathematics.¹ Nor should we overlook the arrival, a few years later in the mid-1990s, of the first symbolic calculator. Wolfram Alpha, introduced in 2009, may have transformed the landscape more than anything before it and now underlies the mathematical computations used by AI systems. We could also add the increasingly frequent use of open-source software such as SageMath and GeoGebra.

Although far less sweeping in scope than AI, several of these systems could represent - and still represent - a threat to mathematics education. The threat consists of disruption on two levels. First, some of these systems were not originally developed for teaching, so deciding to use them as we did could easily lead us astray. Second, many of them became more powerful and easier to use, both in teaching and in research.

The presentation will attempt to answer a question posed by our colleague Guillaume Roy-Fortin,² using mathematical examples closely connected to our teaching. I will try to show that, instead of viewing computer algebra systems as a threat, some of us saw them as an opportunity. This opportunity was put to particularly good use at ÉTS: one need only think of the course notes written by colleagues in the Department of General Education, with exercises requiring the use of technology.

In my humble opinion, however, all this technology must also be used to teach mathematics. If a computer algebra system - and now AI - is used as a black box, or if a symbolic calculator serves almost exclusively as scaffolding, then we will have missed the point. In particular, we will have missed the opportunity to show the rules being used. Showing which rules apply is an excellent way to connect courses with one another and to review concepts, something we too often forget to do with students.

Speaking of rules, a fine example is Albert Rich's symbolic integration system Rubi (Rule-based Integrator). By using pattern recognition to determine which of its integration rules should be applied to a given expression, Rich became convinced that a very large body of mathematical knowledge must be organized systematically in order to identify the precise formula needed to solve a problem. For this reason, and using various software packages, I will present several examples ranging in level from MAT144 to MAT805, in an effort to convince the audience that mathematics teachers still have something important to contribute, even when surrounded by all these tools.

¹ Bruno Buchberger's 2002 article Computer Algebra: The End of Mathematics? is particularly revealing.

² What Future for Automated Mathematics? was the title of the first of Guillaume's talks.